



**MALLA REDDY UNIVERSITY**

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G.O.Ms.No. 14, Higher Education (UE) Department)

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## **Driver Drowsiness Detection**

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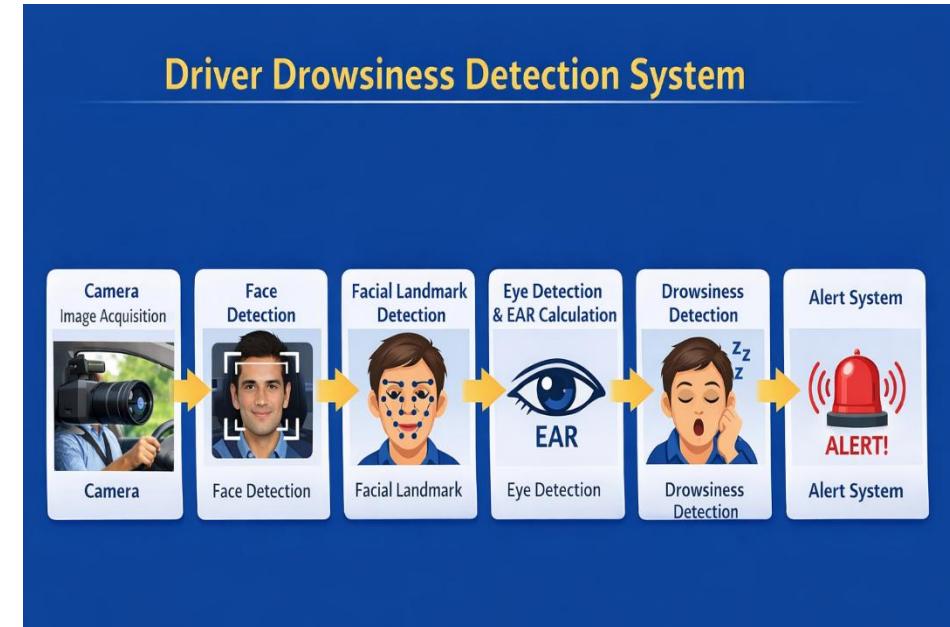
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Branch / Section: DS / Group - III

# OVERVIEW:

- Driver drowsiness is one of the major causes of road accidents worldwide.
- Long driving hours, lack of sleep, and fatigue can reduce driver alertness.
- The system monitors the driver's face using a camera to detect signs of drowsiness.
- It uses **computer vision and machine learning techniques**.
- When drowsiness is detected, the system alerts the driver with an **alarm sound**.
- The system works in **real-time** to prevent accidents.



# ABSTRACT:

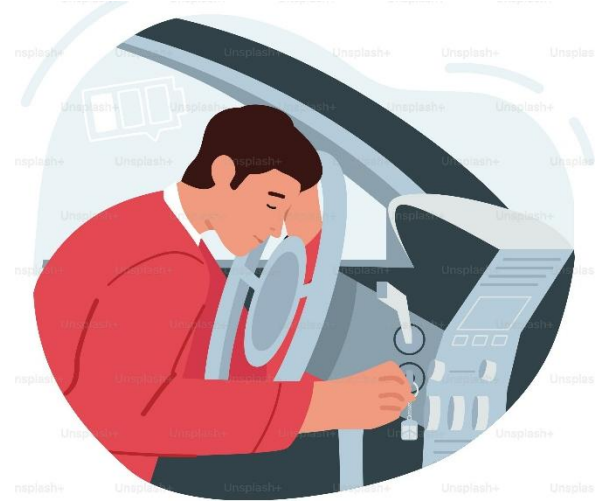
- Driver fatigue is a serious problem that leads to many road accidents.
- The proposed system detects driver drowsiness using **facial feature analysis**.
- A camera continuously captures the driver's face and analyzes eye movement.
- If the eyes remain closed for a certain time, the system identifies the driver as drowsy.
- The system then generates an **alert to wake the driver**.
- This system improves road safety and reduces accident risk.

# OBJECTIVES:

- Detect driver drowsiness in **real-time**.
- Monitor **eye blinking and eye closure**.
- Alert the driver when drowsiness is detected.
- Reduce road accidents caused by fatigue.
- Develop a **low-cost and efficient monitoring system**.
- Improve overall **driver safety**.

# INTRODUCTION:

- Road accidents due to driver fatigue are increasing every year.
- Drivers often become sleepy during long journeys.
- Traditional systems rely on manual monitoring which is not reliable.
- Modern technology allows **automatic detection using computer vision**.
- The driver's facial features such as **eyes, mouth, and head movement** are analyzed.
- The system uses a **camera and machine learning model** to monitor the driver continuously.



# EXISTING SYSTEM:

- Traditional systems depend on **driver self-awareness**.
- Some vehicles use **lane detection sensors**.
- Steering movement monitoring is also used in some cars.
- These systems do not directly monitor the driver's physical condition.
- They are **less accurate in detecting fatigue**.
- Many systems are expensive and not affordable.

## **Limitations**

Not fully reliable.

Cannot detect early drowsiness.

High cost in advanced vehicles.

# PROPOSED SYSTEM:

- The proposed system uses **computer vision and machine learning**.
- A **web camera** continuously captures the driver's face.
- Facial landmarks are detected to analyze **eye blinking patterns**.
- Eye Aspect Ratio (EAR) is calculated to detect eye closure.
- If the driver's eyes remain closed for a certain time, the system identifies drowsiness.
- An **alarm alert** is triggered to wake the driver.

## Advantages

- Real-time detection.
- High accuracy.
- Low cost.
- Easy to implement.

# METHODOLOGY:

## 1. Image Capture

- Camera captures driver's face continuously.

## 2. Face Detection

- Detect the driver's face using **OpenCV**.

## 3. Facial Landmark Detection

- Detect eye landmarks using **Dlib library**.

## 4. Feature Extraction

- Extract eye region and calculate **Eye Aspect Ratio (EAR)**.

## 5. Drowsiness Detection

- If EAR falls below a threshold for a few seconds, drowsiness is detected.

## 6. Alert System

- Alarm sound alerts the driver.



# SYSTEM REQUIREMENTS:

## **Hardware Requirements**

- Computer/Laptop
- Webcam or Camera
- Minimum 4GB RAM
- Processor i3 or above

## **Software Requirements**

- Python
- OpenCV
- Dlib Library
- NumPy
- SciPy
- Operating System (Windows/Linux)

# SYSTEM ARCHITECTURE:

## 1.Camera Module

- Captures driver's face image.

## 2.Face Detection Module

- Detects face from the captured frame.

## 3.Facial Landmark Detection

- Identifies eyes and facial points.

## 4.Eye Aspect Ratio Calculation

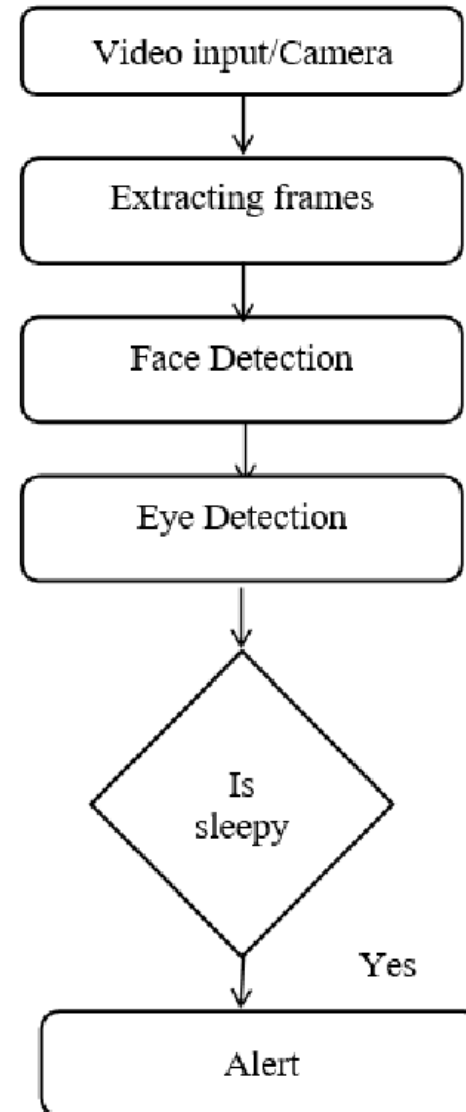
- Calculates EAR to detect eye closure.

## 5.Drowsiness Detection Module

- Checks if the driver is sleepy.

## 6.Alert System

- Produces sound alarm.



**THANK YOU**